

6: Results for a sphere, a circular disk and a vintage car

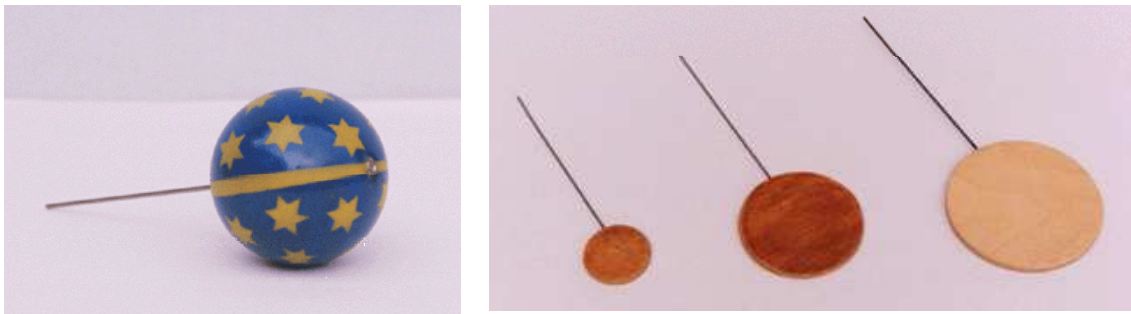


Figure 6.1: For a sphere we use a child's ball of diameter $d = 62.5\text{ mm}$ and with a smooth and shiny surface. Several circular disks go with the merry-go-round. The one considered below has the diameter $d = 76.0\text{ mm}$ and is cut out of 4 mm plywood. These models are mounted on a piece of $2,0\text{ mm}$ spoke from a bicycle wheel.

The experimental results used as an example in table 4.1 are obtained from runs with and without the ball in fig. 6.1. Hence, for the smoothly surfaced sphere with the diameter $d = 62.5\text{ mm}$, the merry-go-round gives the graph of the function $C_d(v)$ shown in fig. 6.2.

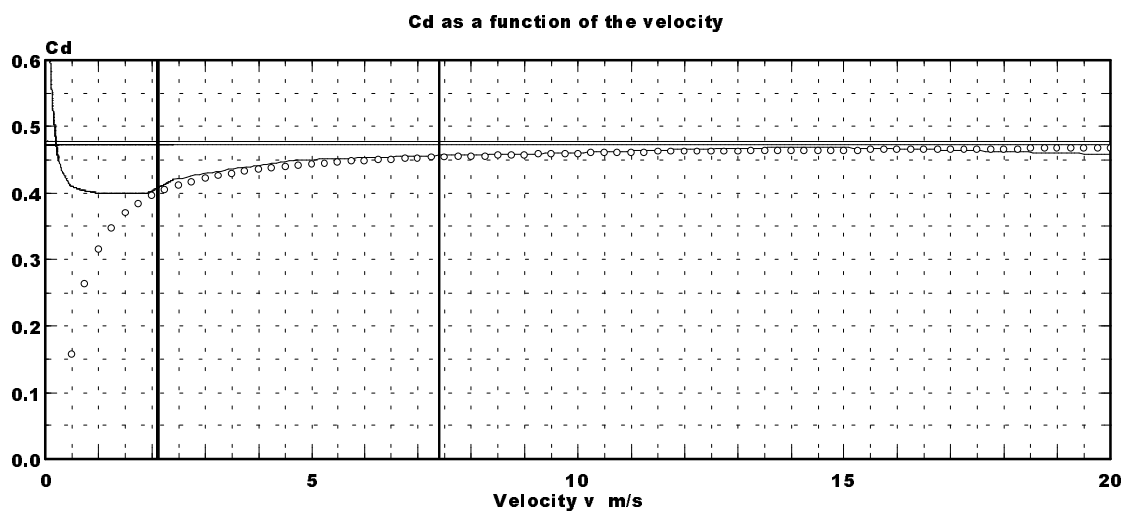


Figure 6.2: The drag coefficient $C_d(v)$ as a function of the velocity v for a smoothly surfaced sphere of diameter $d = 62.5\text{ mm}$, The small circles indicate the graph for

$$C_d(v) = \frac{2F}{\rho A v^2} = \frac{2(\Delta f_1 \cdot v + \Delta f_2 \cdot v^2)}{\rho A v^2} = \frac{c_1}{v} + c_2 = -\frac{0.128 \frac{\text{m}}{\text{s}}}{v} + 0.470,$$

obtained from the experimental results collected in table 4.1 ($\rho = 1.21\text{ kg/m}^3$). The horizontal double line indicates the value of the constant c_2 which $C_d(v)$ approaches in the limit $v \rightarrow \infty$. The two heavy vertical lines indicate the interval of velocities in which the measurements have been carried out. The solid curve indicates what is expected from values in generally accepted tables (see more about this in sec. 8). Note the strong deviation between the two curves *before* the interval. *Within* the interval, the agreement is very good and it seems to be so also *after* the interval. However, as one can see from the next figure, the latter agreement does not last.

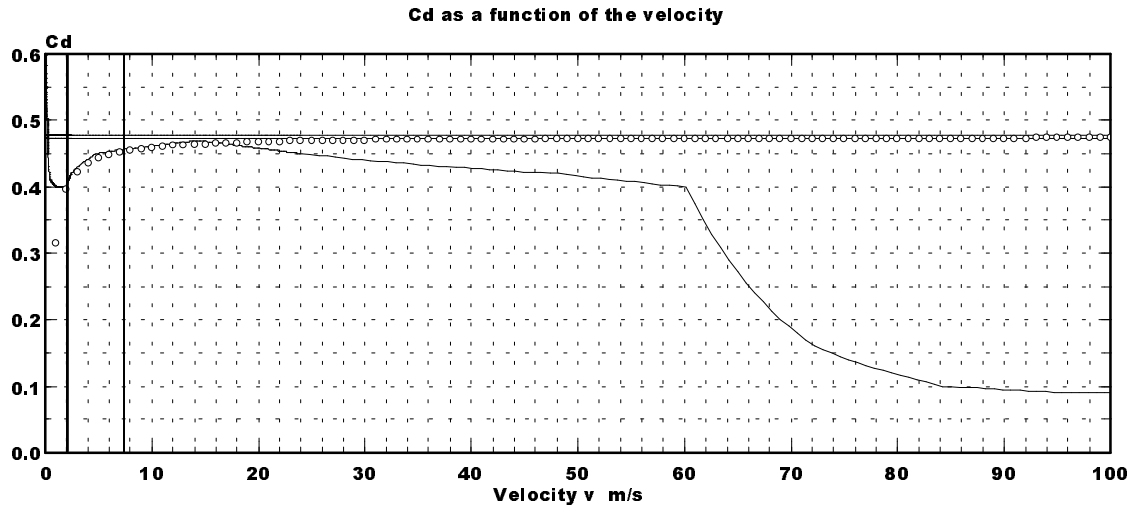


Figure 6.3: The same graphs as in fig. 6.2 but with a much longer interval for the velocity. As one sees, our experimental graph starts to deviate from the correct one when v becomes larger than 20 m/s. At 60 m/s, the correct graph drops rather abruptly. Later it rises somewhat again (compare with fig. 8.1.)

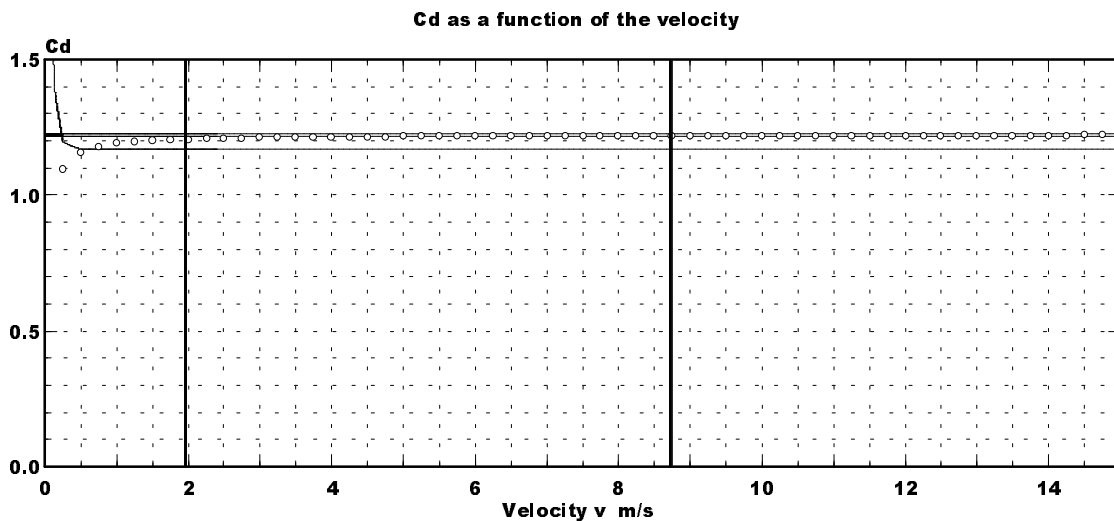


Figure 6.4: The same as in fig. 6.2, but for a circular disk with diameter $d = 76.0 \text{ mm}$. That is, the small circles indicate the results measured by the merry-go-round,

$$C_d(v) = \frac{c_1}{v} + c_2 = -\frac{0.008 \frac{\text{m}}{\text{s}}}{v} + 1.219$$

This time there is no reason for a figure like 6.3: The solid line formed by the table values does *not* suddenly drop down as for the sphere. It goes on forming a horizontal line.

In the next figure we give the results for a rather irregular body - a vintage car.

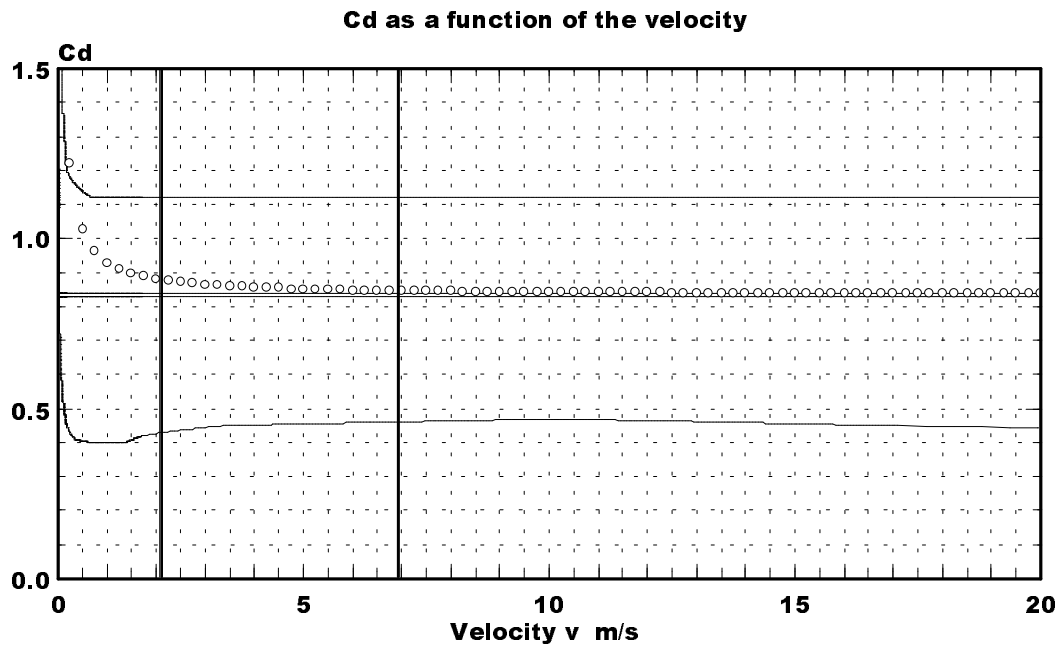


Figure 6.4: The same as in fig. 6.2, but with a 1:18 model of the vintage car, the Mercedes SSK 1928 shown in fig. 7.1, instead of the sphere. This time we get

$$C_d(v) = \frac{c_1}{v} + c_2 = \frac{0.097 \frac{\text{m}}{\text{s}}}{v} + 0.832$$

The two other curves are drawn just for comparison. They show what we should expect if the model had been a circular disk with the same frontal area as the car and if it had been a sphere with that area. If, on the background of this figure, a single constant value of C_d should be given, it should of course be the value $c_2 = 0,832$.

Which experimental factors are important for the accuracy of the c_2 -values

$$\begin{array}{ll} \text{Smooth sphere:} & c_2 = 0,470, \\ \text{Circular disk:} & c_2 = 1,219, \\ \text{Vintage car:} & c_2 = 0,832, \end{array} \quad (6.1)$$

found in this section? Although the fits considered in figs. 3.3 and 3.4 are extraordinarily accurate, the last digit in the c_2 's is hardly reliable. We desist from a proper analysis of the uncertainty which is likely to be fairly difficult. Note however that there are certain factors which do *not* contribute to an error even though one might think so. See *A note on the accuracy ...* at the end of sec. 3.